mas-est Guide

Prerequisites · Install · Upgrade · What gets installed · Log in with the demo users · Connection details · Connect S3 to Manage · Connect SMTP · Day-2 operations · Uninstall

When something fails, see Troubleshooting. What the toolkit doesn't do is in Known limitations.

Placeholders: `<instance>` is your MAS instance ID, `<cluster-domain>` the cluster's apps domain, `<workspace>` the MAS workspace ID.

Prerequisites

Need	Detail
Local tools	`podman`, `oc`, bash 3.2 or later
Cluster access	`oc` logged in as cluster-admin. The install creates namespaces, ClusterRoles and OLM resources.
MAS API key	Name and value, with the **userAdmin** and **systemAdmin** permissions. Preflight rejects a key without them.
Storage	A block (RBD) storage class. Preflight recommends one.
MAS version	9.0.x or 9.1.x. On MAS 9.0, leave out OIDC. MAS 9.2 has not been tested yet.

Install

1. Bootstrap the CLI

```
export MAS_EST_IMAGE='quay.io/lee_forster/mas-external-services-tool:v0.1.16'
mkdir -p "$HOME/mas-est"
podman run -ti --rm -v "$HOME/mas-est:/tmp" --pull always "$MAS_EST_IMAGE" bootstrap
 ↪  --force
export PATH="$HOME/mas-est:$PATH"
mas-est version
```

`version` must print `0.1.13`. `--force` overwrites an earlier runtime and is safe on a first install.

2. Preflight

```
mas-est preflight
```

Enter the MAS SCIM URL, `https://api.<instance>.<cluster-domain>/scim/v2`, and the API key. Success looks like this:

```
[ok] oc: oc CLI found on PATH
[ok] cluster-login: logged in as kube:admin against https://api.<cluster-
domain>:6443
[ok] storage-classes: discovered 5 storage classes; recommended=localblock
[warn] storage-selection: default storage class ocs-storagecluster-cephfs looks filesystem-
backed; preferring localblock for PostgreSQL
```

```
[ok] mas-base-url: parsed MAS host api.<instance>.<cluster-domain>
[ok] mas-api-key: MAS API key <token-name> authenticated and is authorized for the SCIM API
[ok] mas-route-lookup: matched route mas-<instance>-core/<instance>-api
```

`install` runs two more checks before it starts: whether MAS exposes the OIDC API, and whether login providers named `default` already exist.

3. Install

```
mas-est install
```

Prompt	What to answer
Products to install	LDAP directory, Keycloak identity provider, SCIM bridge, S3 object storage, SMTP Mailpit capture server. SCIM adds Keycloak and LDAP; Keycloak adds LDAP.
MAS SCIM base URL, MAS API token name, MAS API token value	Same as preflight
Workspace ID, MAS profile ID	The workspace is filled in when MAS has one. Profile ID defaults to `demo`.
MAS instance ID, MAS core namespace	Filled in from the SCIM URL
Configure MAS authentication providers?	Yes. Then choose `ldap`, `oidc`, `saml`. On MAS 9.0 leave out `oidc`.
Primary storage class, SCIM bridge storage class	The block class preflight recommended

Answers marked `[derived]` were worked out for you. Confirm at `Proceed with install?`.

The install runs as a Job named `mas-est-install` in the `mas-est` namespace and streams its log to your terminal, reconnecting if the connection drops. Ctrl-C detaches from the log. The Job keeps running.

```
mas-est logs --component install-job --follow   # reattach
oc delete job mas-est-install -n mas-est        # cancel, or clear a finished Job before
 ↪  re-running
mas-est install --local                         # run on this machine instead of in the
 ↪  cluster
```

Success ends with the summary:

```
Install summary (total 17m27s)
  ✓ Install LDAP, Keycloak, SCIM bridge      (5m44s)
  ✓ Install MinIO S3                         (1m47s)
  ✓ Install Mailpit SMTP                     (3s)
  ✓ Configure MAS LDAP / OIDC / SAML         (9m53s)
```

The Job's ServiceAccount gets a ClusterRole close to cluster-admin. The install reads secrets in the MAS namespaces and runs `pods/exec` on MongoDB to link the SCIM users. The permissions are applied by the CLI you run, not by the Job image, so re-bootstrap before installing after an upgrade. `--local` needs the same cluster-admin rights, and preflight's `cluster-permissions` check names anything missing. The MAS API calls honour `HTTPS_PROXY` and `NO_PROXY`.

Scripted install, with every value as a flag:

```
mas-est install --non-interactive \
  --components ldap,keycloak,scim,s3,smtp \
  --mas-base-url 'https://api.<instance>.<cluster-domain>/scim/v2' \
  --mas-api-token-name '<name>' --mas-api-token-value '<value>' \
  --workspace-id '<workspace>' --mas-instance-id '<instance>' \
  --configure-mas-auth --mas-auth-providers ldap,oidc,saml \
  --storage-class '<block-class>' --scim-bridge-storage-class '<block-class>'
```

4. Verify

```
mas-est status
```

Every deployment shows `ready=1/1`, the CSV shows `phase=Succeeded`, and every PVC shows `phase=Bound`. Then log in as one of the demo users.

Upgrade

```
mas-est update
export MAS_EST_IMAGE='quay.io/lee_forster/mas-external-services-tool:v<new-version>'
mas-est version
```

`update` looks up the newest release on Quay and re-runs bootstrap with podman (or docker) into the directory that holds `mas-est`. `mas-est update --check` only reports whether a newer release exists, and `--version vX.Y.Z` installs a specific tag. Update `MAS_EST_IMAGE` afterwards as `update` prints: `install` refuses to run while it names a different version than the binary.

Versions before v0.1.14 don't have `update`. Upgrade those by re-running bootstrap with the new tag:

```
export MAS_EST_IMAGE='quay.io/lee_forster/mas-external-services-tool:v<new-version>'
podman run -ti --rm -v "$HOME/mas-est:/tmp" --pull always "$MAS_EST_IMAGE" bootstrap
 ↪  --force
```

Nothing in the cluster changes until you run `mas-est install` again. Re-running `install` or `mas-auth apply` is safe: it keeps the Keycloak OIDC client secret, the OpenLDAP TLS certificates, the self-registration entries for other providers, and the existing storage classes.

Upgrading an install made before v0.1.16

Operator 0.0.16 moves OpenLDAP's configuration onto its volume, so OpenLDAP survives a reschedule. On the first start after the upgrade, an init container rebuilds the configuration and checks that it opens the existing directory, keeping all entries. It writes an LDIF export to `backup-<timestamp>/` on the OpenLDAP volume. If the check fails, the log shows a block of `!!!` lines. OpenLDAP then starts a fresh directory and re-seeds the demo users, and the old database stays in the backup folder.

Installs before v0.1.16 also share one set of Keycloak realm signing keys, which were published in this repository. Rotate them on every existing install:

```
KC=$(oc -n mas-est get pod -l app.kubernetes.io/component=keycloak -o name | head -1)
oc -n mas-est exec -i "$KC" -- bash -s <<'SH'
set -euo pipefail
export HOME=/tmp
k=/opt/keycloak/bin/kcadm.sh
$k config credentials --server http://localhost:8080 --realm master --user
 ↪  "$KC_BOOTSTRAP_ADMIN_USERNAME" --password "$KC_BOOTSTRAP_ADMIN_PASSWORD"
```

```
RID=$($k get realms/maximo --fields id --format csv --noquotes)
for p in rsa-generated rsa-enc-generated hmac-generated aes-generated; do
  $k create components -r maximo -s name="$p-rotated" -s providerId="$p" \
    -s providerType=org.keycloak.keys.KeyProvider -s parentId="$RID" -s
 ↪  'config.priority=["200"]'
done
for id in faa58d00-c7c1-408f-b4c0-f86e898d8008 4ceaa268-6821-46e7-a8a1-777325620fbc \
          0260823f-687d-4917-8276-e935b3b21bfd 6252f6c5-b495-4ad4-9a4c-b1588f13338b; do
  $k delete "components/$id" -r maximo || true
done
$k delete users/170c81a4-3988-47a2-ba5a-c4bc3da16a04 -r maximo || true
SH
mas-est mas-auth apply --mas-instance-id <instance> --self-reg-workspace <workspace>
 ↪  --providers saml
```

Users sign in again afterwards. OIDC picks up the new keys by itself. The last command re-uploads the SAML metadata, because MAS keeps a copy with the old signing certificate. MAS takes up to about 15 minutes to apply it, and the command waits for that. SAML logins fail until then; no restart is needed.

What gets installed

Component	In namespace `mas-est`	In namespace `mas-<instance>-core`
MAS IAM operator	CSV `mas-iam-operator.v0.0.17`, from CatalogSource `mas-iam-operator` in `openshift-marketplace`	
OpenLDAP	`deployment/mas-est-iam-openldap`	`idpcfg/<instance>-ldap-default-system`
Keycloak, realm `maximo`, with PostgreSQL	`deployment/mas-est-iam`, route `mas-est-iam`	`idpcfg/<instance>-oidc-default-system`, `idpcfg/<instance>-saml-default-system`
SCIM bridge	`deployment/scim-bridge`	SCIM profile demo
MinIO	`deployment/mas-minio`, routes `mas-minio-api` and `mas-minio-console`	`objectstoragecfg/ibm-mas-<instance>-objectstoragecfg-system`
Mailpit	`deployment/mas-mailpit`, route `mas-mailpit`	
Self-registration		`configmap/<instance>-selfreg`. Users are created on first login with PREMIUM entitlement and Manage in `<workspace>`.

MAS calls a login provider an IDPCfg. The provider ID is always `default`, which is what makes MAS administration show each provider as configured. If providers with that ID already exist, the install overwrites them. Back them up first:

```
oc get idpcfg -n mas-<instance>-core -o yaml > idpcfg-backup.yaml
```

The install also raises the memory limit of `<instance>-entitymgr-idpcfg` to 2 Gi through the Suite CR, because the MAS default of 512 Mi runs out while the providers are applied. `--idpcfg-memory-limit off` skips this.

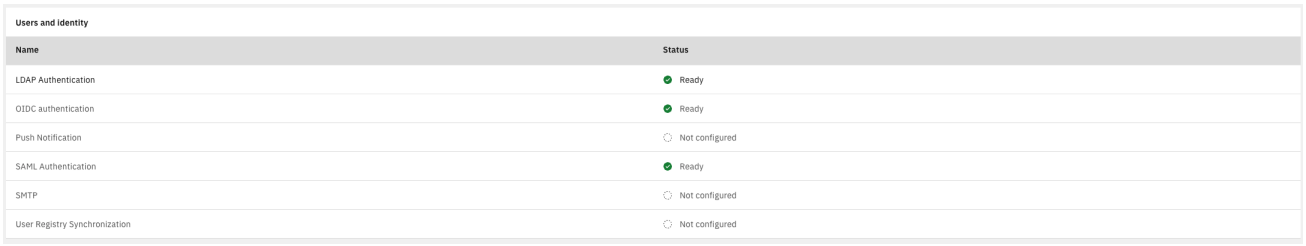


Figure 1: MAS administration, Users and identity, with LDAP, OIDC and SAML authentication all Ready

Log in with the demo users

User	Directory	Log in with	Password
`ldap.user1`, `ldap.user2`	OpenLDAP	LDAP	`mas-est ldap-info --show-user-passwords`
`oidc.user1`, `oidc.user2`	Keycloak	OIDC	`maxadmin`
`saml.user1`, `saml.user2`	Keycloak	SAML	`maxadmin`
`scim.user1`, `scim.user2`	Keycloak	OIDC	`maxadmin`

LDAP, OIDC and SAML users are created in MAS on their first login. SCIM users are created by the SCIM bridge before anyone logs in.

Use a private browser window for every login. Keycloak keeps you signed in after you log out of MAS, so the next login in the same window returns the previous user.

To check the Keycloak logins without a browser, run this from a checkout of this repo (Python 3, no packages):

```
make login-check HOME_URL=https://home.<instance>.<apps-domain>
```

It logs in each OIDC, SAML and SCIM demo user through MAS and passes a user when MAS accepts the login. A failure names the MAS error code, such as `AIUOM0024E` (user deactivated in MAS). LDAP logins aren't covered. A user's first check self-registers it, as a first browser login does.

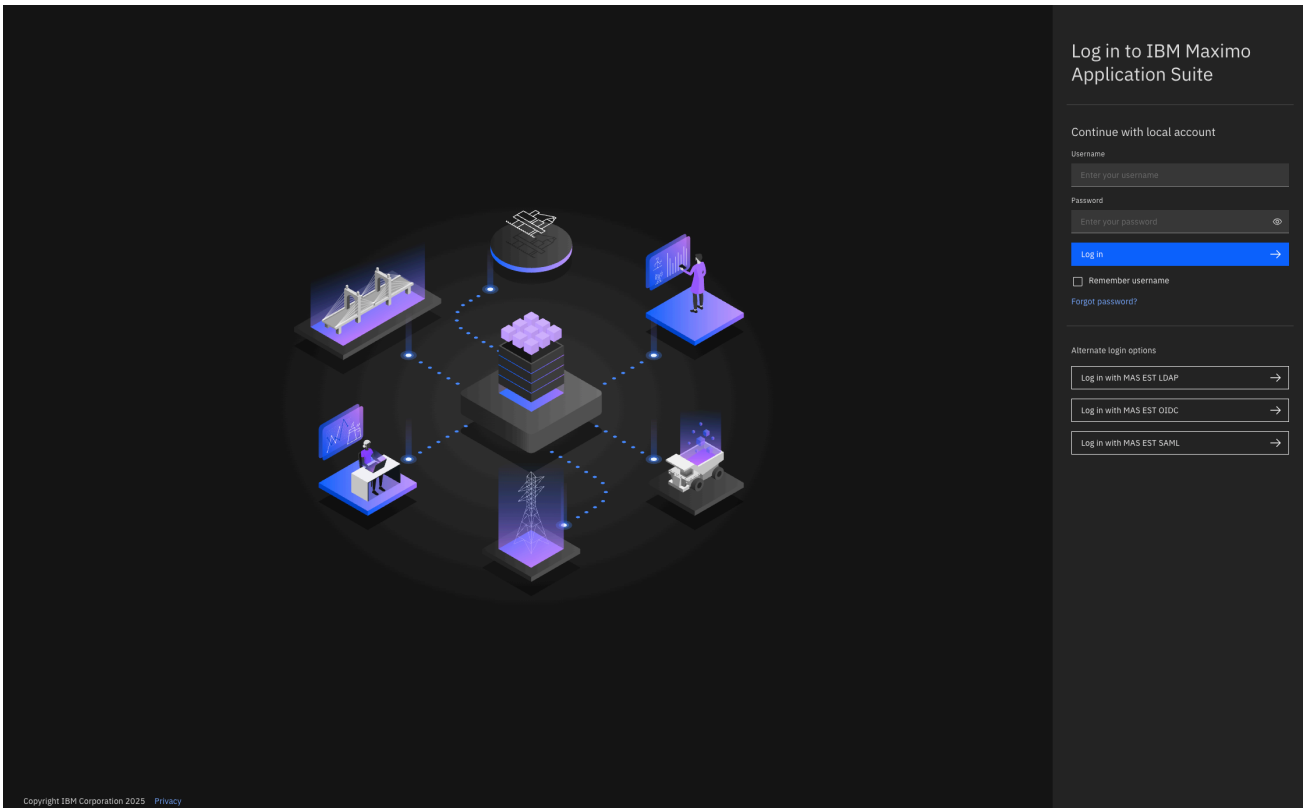


Figure 2: MAS login page with the Log in with MAS EST LDAP, OIDC and SAML options

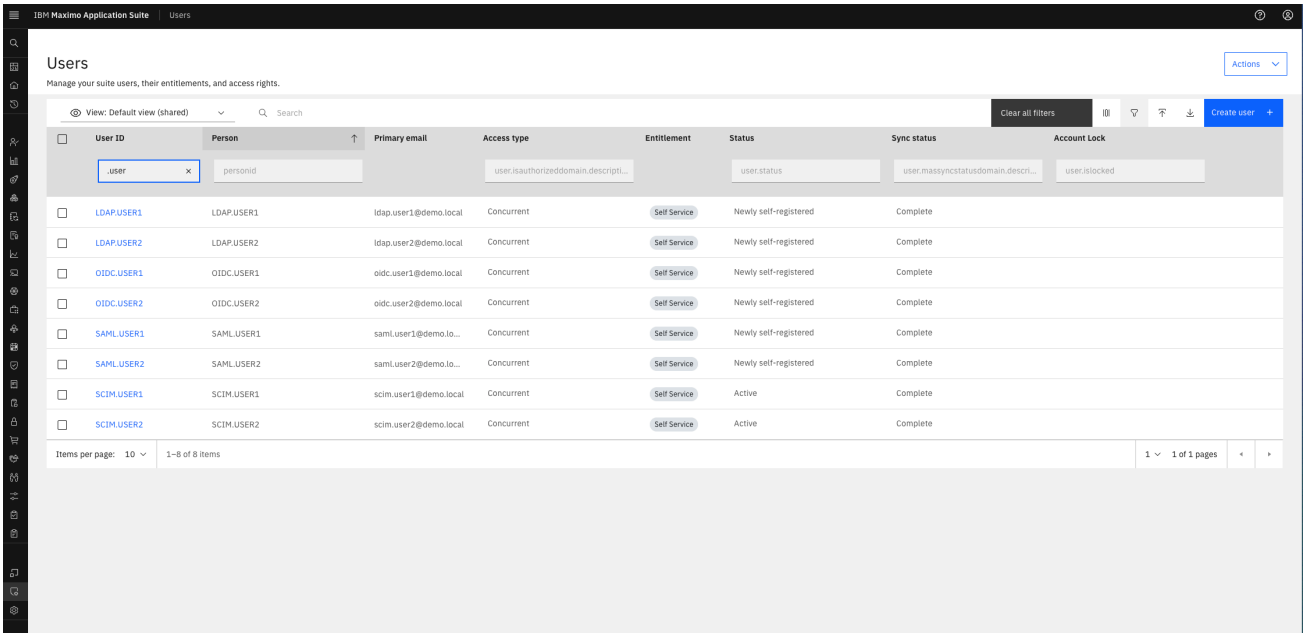


Figure 3: MAS Users after every demo user has logged in once: SCIM users Active, the others Newly self-registered

Add a SCIM user

1. Open the Keycloak admin console at the `mas-est-iam` route, realm `maximo`. The admin password is in secret `mas-est-iam-bootstrap-admin`.
2. Create a user whose username starts with `scim.`, give it an email address, and add it to the group `mas-scim-users`.

3. Wait up to 5 minutes. `mas-est logs` shows `action=create username=scim.<name>` when the bridge has created it in MAS.
4. Link the user to its OIDC identity. Without this step it can't log in or reach Manage.

   `mas-est mas-auth apply --mas-instance-id <instance> --self-reg-workspace <workspace>`

   On MAS 9.0 add `--providers ldap,saml`.

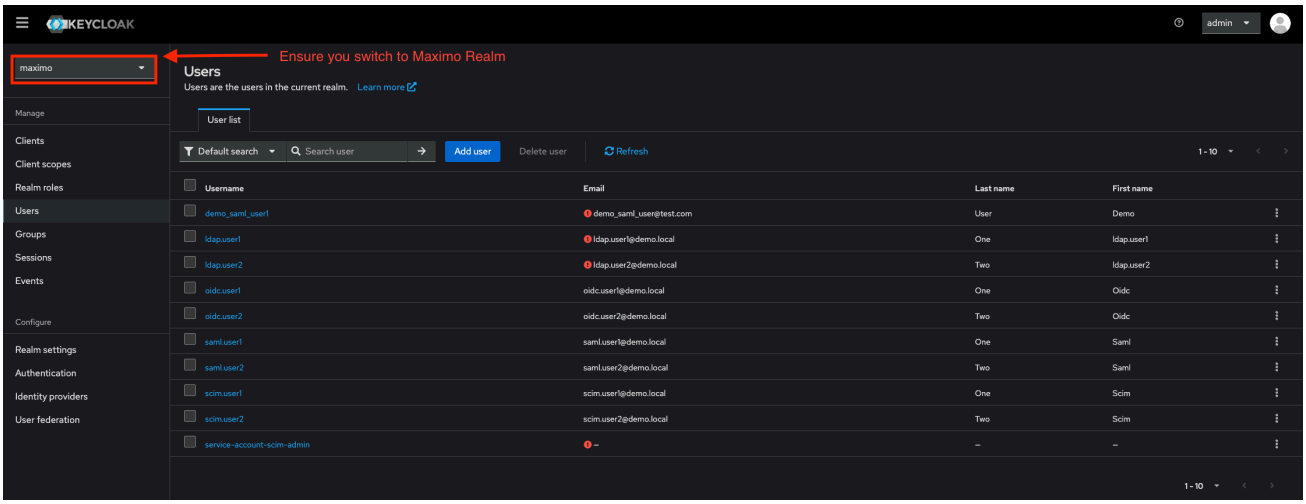


Figure 4: Keycloak admin console, realm maximo, showing the demo users

Connection details

```
mas-est details        # --component ldap|oidc|saml|s3|smtp; --show-secrets
mas-est ldap-info      # --show-password; --show-user-passwords
```

Both print what MAS is configured with. The raw values are in these resources in `mas-est`:

Resource	Keys
`secret/mas-est-ldap-connection`	`url`, `baseDN`, `bindDN`, `bindPassword`, `userIdMap`, and `ca.crt` when the LDAP CA was found
`secret/mas-est-oidc-connection`	`issuerUrl`, `discoveryUrl`, `authorizationEndpoint`, `tokenEndpoint`, `jwksEndpoint`, `clientId`, `clientSecret`, `realm`, `redirectUri`
`secret/mas-est-saml-connection`	`entityId`, `acsUrl`, `sloUrl`, `idpMetadataUrl`, `nameIdFormat`, and `idpMetadata` when the metadata fetch succeeded
`secret/mas-est-s3-connection`	`provider`, `endpoint`, `manageEndpoint`, `externalEndpoint`, `consoleUrl`, `accessKey`, `secretKey`, `region`, `bucket`, `siblingBuckets`
`configmap/mas-est-smtp-connection`	`host`, `port`, `from`, `webUI`, `tls`, `authentication`, `relayEnabled`

```
oc get secret mas-est-oidc-connection -n mas-est -o jsonpath='{.data.clientSecret}' |
 ↪  base64 -d
```

LDAP settings

Setting	Value
URL, inside the cluster	`ldaps://mas-est-iam-openldap.mas-est.svc.cluster.local:636`
Bind DN	`cn=admin,dc=demo,dc=local`. Password: secret `mas-est-iam-openldap-admin`, key `password`
Base DN	`dc=demo,dc=local`
Users, groups	`ou=users,dc=demo,dc=local`, `ou=groups,dc=demo,dc=local`
User attribute	`uid`
Group object class, member attribute	`groupOfUniqueNames`, `uniqueMember`

From your machine: `oc -n mas-est port-forward svc/mas-est-iam-openldap 1636:636`.

Connect S3 to Manage

The install connects MinIO to MAS Suite through the `ObjectStorageCfg`. Manage attachments, called doclinks, need these steps.

1. In Manage, open **System Properties** and set:

Property	Value
`mxe.cosendpointuri`	`http://mas-est.svc.cluster.local:9000`
`mxe.cosbucketname`	`mas-s3-demo`
`mxe.cosregion`	`us-east-1`. Add it with **New Row** if it doesn't exist.
`mxe.cosaccesskey`	`minioadmin`
`mxe.cossecretkey`	Secret `mas-minio-root`, key `MINIO_ROOT_PASSWORD`
`mxe.attachmentstorage`	The class name in the block below
`mxe.doclink.doctypes.defpath`	`cos:doclinks`
`mxe.doclink.doctypes.topLevelPaths`	`cos:doclinks`
`mxe.doclink.path01`	`cos:doclinks=<manage-ui-base-url>`
`mxe.doclink.securedAttachment`	`True`

   Value for `mxe.attachmentstorage`:

   ```
   com.ibm.tivoli.maximo.oslc.provider.COSAttachmentStorage
   ```

2. Click **Live Refresh**.
3. In **Document Types**, set each document type's default path to `cos:doclinks/<name>`, for example `cos:doclinks/attachment`.

4. Restart Manage:

```
oc rollout restart deployment/<instance>-<workspace>-all -n mas-<instance>-manage
```

Use the endpoint above, not the MinIO route. Manage addresses buckets as `<bucket>.<host>`. The install creates a Service for each bucket name that resolves only inside the cluster, and the route returns HTTP 503 or certificate errors. Without `mxe.cosregion`, every request fails with `SignatureDoesNotMatch`.

Browse the buckets at the `mas-minio-console` route as user `minioadmin`, with the `MINIO_ROOT_PASSWORD` key of secret `mas-minio-root`.

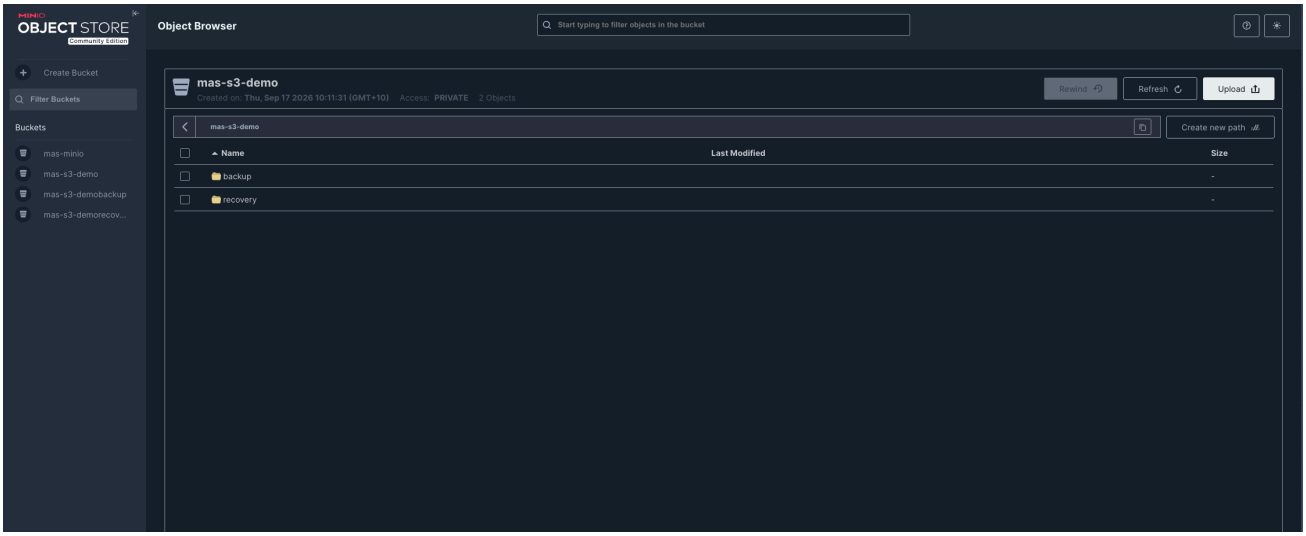


Figure 5: MinIO console showing bucket mas-s3-demo with its backup and recovery folders, and the sibling buckets in the sidebar

Connect SMTP

Mailpit captures every message and shows it at the `mas-mailpit` route. It delivers nothing unless you add a relay. The install doesn't point MAS at it. Set these in MAS administration:

Setting	Value
Host	`mas-mailpit.mas-est.svc.cluster.local`
Port	`1025`
TLS	Off
Authentication	None

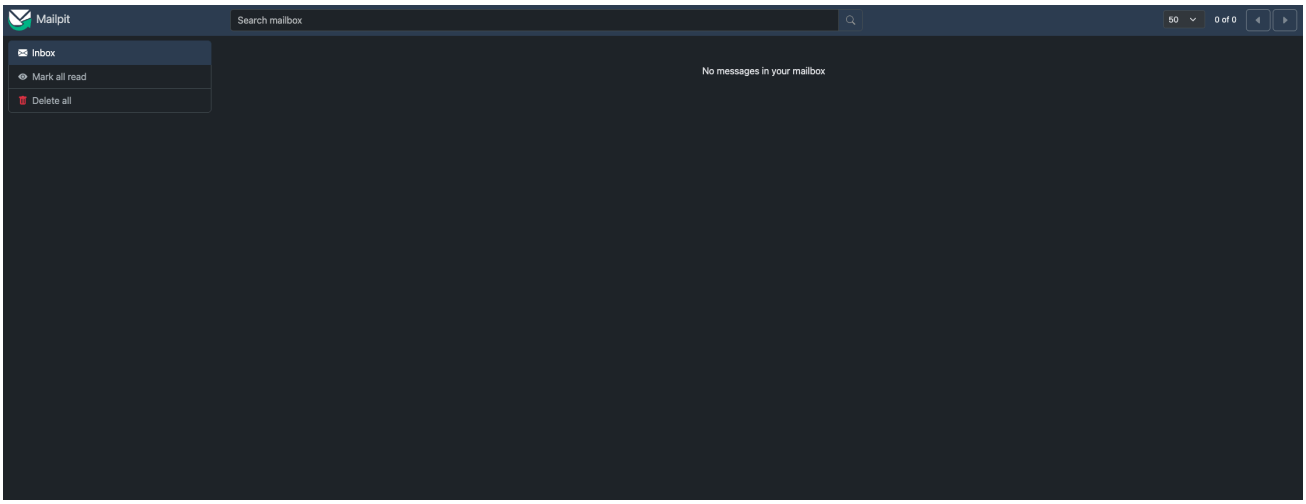


Figure 6: Mailpit inbox at the mas-mailpit route

To forward captured mail to real inboxes, install with relay flags. They also work with `mas-est smtp install-mailpit`.

```
--smtp-relay-host smtp.gmail.com --smtp-relay-port 587 --smtp-relay-starttls \
--smtp-relay-username 'lab@example.com' --smtp-relay-password "$APP_PASSWORD" \
--smtp-relay-from 'lab@example.com'
```

The MAS settings stay the same. Gmail needs an App Password. For SendGrid the username is the literal `apikey`. Microsoft 365 needs OAuth, which Mailpit doesn't support.

Day-2 operations

Health and logs

```
mas-est status
mas-est logs --component bridge --tail 300    # --follow to stream
```

Log components: `operator`, `keycloak`, `openldap`, `bridge`, `profile-bootstrap`, `minio`, `minio-init`, `smtp`, `install-job`.

SCIM bridge settings

The bridge reads `configmap/scim-bridge-config` and `secret/scim-bridge-secret` at startup. These commands restart it for you:

```
mas-est config view
mas-est config set mas-api-token --token-name '<name>' --token-value '<value>'   # rotate
 ↪  the MAS API key
mas-est config set bridge --log-level debug                                      # or
 ↪  --payload-logging true
mas-est restart bridge                                                           # after
 ↪  editing the ConfigMap by hand
```

The bridge syncs a user only if it is in the Keycloak group named by `SCIM_BRIDGE_INCLUDE_GROUPS` (`mas-scim-users`) and its username starts with `SCIM_BRIDGE_INCLUDE_USERNAME_PREFIX` (`scim.`). The full setting list is in services/scim-bridge/README.md.

Payload logging writes usernames, names and email addresses to the log. Turn it off when you're done.

From bridge v0.1.3, MAS errors don't stop a user syncing. Temporary errors retry with backoff, from 30 seconds up to 30 minutes. A user MAS rejects three times with a 400, 404 or 422 is parked until the Keycloak user, its profile or the bridge config changes. To retry a parked user, change anything on it in Keycloak. The bridge serves `/healthz`, `/readyz` and `/status` on port 8081; `/status` reports the last successful cycle and how many users are retrying or parked.

Re-apply or remove the MAS login providers

```
mas-est mas-auth apply --mas-instance-id <instance> --self-reg-workspace <workspace>   #
 ↪  re-create providers and re-link SCIM users
mas-est mas-auth delete --mas-instance-id <instance>                                    #
 ↪  remove all three; --providers to pick
```

`apply` reads the API key from `secret/scim-bridge-secret`. Always pass `--self-reg-workspace`, or self-registration falls back to a workspace named `workspace`. `--providers` limits the run to those providers and leaves the others' self-registration entries in place. If MongoDB isn't in namespace `mongoce`, pass `--mongo-namespace`.

Report a problem

```
mas-est support-bundle
```

Writes a `mas-est-support-mas-est-<timestamp>` directory with status, events, pod and route listings, ConfigMaps and component logs. No Secret values are written. Check it for customer hostnames before sharing it outside the team.

Uninstall

```
mas-est uninstall                          # also deletes SCIM profile demo
mas-est uninstall --skip-profile-delete    # keeps the profile
```

Deletes the `mas-est` namespace. It leaves behind:

- The login providers. Remove them with `mas-est mas-auth delete --mas-instance-id <instance>`.
- `configmap/<instance>-selfreg` and the Suite CR memory bump.
- MAS user records in MongoDB and Manage.
- `clusterrole/mas-est-installer`, `clusterrolebinding/mas-est-installer-mas-est`, the SCC `mas-est-iam-openldap-tls-generator`, and the CRD `masiamstacks.iam.mas.ibm.com`.
- CatalogSource `mas-iam-operator` in `openshift-marketplace`.

Reinstalling onto the same MAS with leftover user records can break self-registration and Manage sync. The fixes are in Troubleshooting.